Adopted Levels

Type	Author	History Citation	Literature Cutoff Date
Full Evaluation	Balraj Singh	NDS 156, 148 (2019)	31-Jan-2019

Q(β^-)=−4730 *SY*; S(n)=7100 *SY*; S(p)=2490 *SY*; Q(α)=10072 *13* 2017Wa10
Estimated uncertainties (2017Wa10): 970 for Q(β^-), 1010 for S(n), 1080 for S(p).
Q(α)=10.080 MeV *20* (2017Ka66), 10.074 MeV *15* (2016Ho09), 10.07 MeV *3* (2015Og05,2017Og01).
S(2n)=12870 *1040* (syst,2017Wa10). S(2p)=6450 (theory,1997Mo25).
EVR=Evaporation Residues.

The ^{288}Fl nuclide has been identified by Dubna-Livermore-ORNL collaboration (2004Og12, also many other papers from this group), and later independently by the GSI group using TASCA separator (2012Ho12). A total of 27 EVR-α-SF correlated decay chains have been reported. Additionally, ^{288}Fl is also produced as alpha-daughter of ^{292}Lv, for which ten EVR-α-α-SF correlated decay chains have been reported (see ^{292}Lv Adopted Levels).

Experiments at FLNR-JINR-Dubna, using U-400 cyclotron and Dubna gas-filled recoil separator (DGFRS):
^{288}Fl nuclide produced directly in experiments using nuclear reactions at FLNR-JINR-Dubna, and UNILAC-GSI laboratories. The analyses of the experiments at Dubna were also done independently at the Lawrence Livermore National Laboratory (LLNL). The radioactive targets were fabricated at the Oak Ridge National Laboratory (ORNL).

2017Ka66 (also 2014MoZV): three correlated events were reported in ^{248}Cm(^{48}Ca,4n),E=261.6 MeV reaction using RIKEN Linear Accelerator (RILAC) and gas-filled recoil ion separator (GARIS). Two EVR-α-α-SF correlated events and the following one tentative EVR-α-α-α-SF event were observed, all starting from ^{292}Lv, with production σ=3.1 pb *+28−18*. All three events produced ^{288}Fl. See ^{292}Lv Adopted Levels for details.

2004Og07, 2004Og12 (also 2004OgZZ): ^{288}Fl formed in ^{244}Pu(^{48}Ca,4n),E=243,250,257 MeV (E*=41,47,51 MeV) at Dubna. Measured EVR-α-SF events, $T_{1/2}$, Eα, production σ, excitation functions. Based on their analysis of events assigned to ^{289}Fl in the present work, 2004Og07 point out that two events assigned to ^{288}Fl in their earlier studies (2000Og05,2000Og07,2001Og01,2001Og02, 2001Og06,2002Og03,2002Og09,2002Og13) belonged to ^{289}Fl, instead.
History of 12 EVR-α-SF correlated events (2004Og07), where Δt values are time differences between successive decays.
Event #1: Energy of the evaporation residue (EVR)=7.7 MeV: 2004Og07.
$E_{\alpha 1}$=9.98 MeV *8*, Δt1=0.826 s, assigned to ^{288}Fl.
E_{SF}=229 MeV, Δt_2=0.177 s, assigned to ^{284}Cn.
^{284}Cn decays by SF.
Event #2: Energy of the EVR=11.9 MeV: 2004Og07.
$E_{\alpha 1}$=2.9 MeV, Δt_1=0.838 s, assigned to ^{288}Fl.
E_{SF}=225 MeV, Δt_2=0.168 s, assigned to ^{284}Cn.
^{284}Cn decays by SF.
Event #3: Energy of the EVR=9.8 MeV: 2004Og07.
$E_{\alpha 1}$ missed.
E_{SF}=191 MeV, Δt_2=0.565 s, assigned to ^{284}Cn.
^{284}Cn decays by SF.
Event #4: Energy of the EVR=12.2 MeV: 2004Og07.
$E_{\alpha 1}$=10.02 MeV *7*, Δt_1=0.378 s, assigned to ^{288}Fl.
E_{SF}=137 MeV, Δt2=0.326 s, assigned to ^{284}Cn.
^{284}Cn decays by SF.
Event #5: Energy of the EVR=11.5 MeV: 2004Og07.
$E_{\alpha 1}$=9.97 MeV *7*, Δt1=0.427 s, assigned to ^{288}Fl.
E_{SF}=230 MeV, Δt_2=0.072 s, assigned to ^{284}Cn.
^{284}Cn decays by SF.
Event #6: Energy of the EVR=7.4 MeV: 2004Og07.
$E_{\alpha 1}$=9.91 MeV *9*, Δt_1=0.676 s, assigned to ^{288}Fl.
E_{SF}=150 MeV, Δt_2=0.007 s, assigned to ^{284}Cn.

Adopted Levels (continued)

^{284}Cn decays by SF.
Event #7: Energy of the EVR=11.5 MeV: 2004Og07.
$E_{\alpha 1}$=9.92 MeV *8*, Δt_1=0.083 s, assigned to ^{288}Fl.
E_{SF}=221 MeV, Δt_2=0.028 s, assigned to ^{284}Cn.
^{284}Cn decays by SF.
Event #8: Energy of the EVR=12.1 MeV: 2004Og07.
$E_{\alpha 1}$=9.93 MeV *8*, Δt_1=0.188 s, assigned to ^{288}Fl.
E_{SF}=205 MeV, Δt_2=0.130 s, assigned to ^{284}Cn.
^{284}Cn decays by SF.
Event #9: Energy of the EVR=11.8 MeV: 2004Og07.
$E_{\alpha 1}$=9.9 MeV *6*, Δt_1=0.930 s, assigned to ^{288}Fl.
E_{SF}=155 MeV, Δt_2=0.069 s, assigned to ^{284}Cn.
^{284}Cn decays by SF.
Event #10: Energy of the EVR=11.2 MeV: 2004Og07.
$E_{\alpha 1}$=9.94 MeV *7*, Δt_1=0.391 s, assigned to ^{288}Fl.
E_{SF}=221 MeV, Δt_2=0.389 s, assigned to ^{284}Cn.
^{284}Cn decays by SF.
Event #11: Energy of the EVR=11.7 MeV: 2004Og07.
$E_{\alpha 1}$=9.87 MeV *9*, Δt_1=0.823 s, assigned to ^{288}Fl.
E_{SF}=194 MeV, Δt_2=0.063 s, assigned to ^{284}Cn.
^{284}Cn decays by SF.
Event #12: Energy of the EVR=11.0 MeV: 2004Og07.
$E_{\alpha 1}$=10.05 MeV *19*, Δt_1=4.468 s, assigned to ^{288}Fl.
E_{SF}=199 MeV, Δt_2=0.123 s, assigned to ^{284}Cn.
^{284}Cn decays by SF.
Event #13: Energy of the EVR=9.5 MeV: 2004Og12.
$E_{\alpha 1}$=9.86 MeV *45*, Δt_1=3.852 s, assigned to ^{288}Fl.
E_{SF}=202 MeV, Δt_2=0.202 s, assigned to ^{284}Cn.
^{284}Cn decays by SF.

2010Ei01: ^{244}Pu(^{48}Ca,4n),E=233-251 MeV at Dubna. Investigated deposition pattern of Fl on a gold surface using thermochromatography technique and using the COLD apparatus, and found that Fl is as volatile as simultaneously studied elements Hg, At, and Cn. Two correlated events were assigned to ^{288}Fl.
Event #1: Energy of the evaporation residue (EVR)=9.95 MeV: 2010Ei01.
$E_{\alpha 1}$=9.95 MeV, assigned to ^{288}Fl.
E_{SF}=62 MeV, Δt_2=0.109 s, assigned to ^{284}Cn.
^{284}Cn decays by SF.
Event #2: Energy of the EVR=9.81 MeV: 2010Ei01.
$E_{\alpha 1}$=9.81 MeV, assigned to ^{288}Fl.
E_{SF}=108 MeV, Δt_2=0.104 s, assigned to ^{284}Cn.
^{284}Cn decays by SF.

Experiments at GSI, using universal linear accelerator (UNILAC), Transactinide Separator and Chemistry Apparatus (TASCA) and Cryo-Online Multidetector for Physics And Chemistry of Transactinides (COMPACT):

2010Du06, 2011Ga19: ^{244}Pu(^{48}Ca,4n),E=259.4, 254.6 MeV; beam extracted from an electron cyclotron resonance (ECR) ion source and accelerated by the universal linear accelerator (UNILAC) at the GSI. Target=438 μg/cm^2 ^{244}Pu enriched to 97.9% purity backed with a 2.22 μm thick Ti foil. Product and beam nuclei were separated using the Transactinide Separator and Chemistry Apparatus (TASCA) based on magnetic rigidity. Evaporation residues were implanted into the focal plane detector consisting of two 300 μm thick double sided silicon strip detectors (DSSSDs) after passing through a multiwire proportional counter (MWPC). Two single sided silicon strip detectors (SSSSDs) mounted behind the DSSSDs to suppress light, low-ionizing particle detection and an additional eight SSSSDs were mounted upstream in a five-sided box configuration to detect alpha particles and SF fragments escaping the DSSSDs. A Ge cluster detector with seven Ge crystals was mounted behind the Al backplate of the

Adopted Levels (continued)

TASCA detection chamber for $\alpha\gamma$ coincidences. Measured EVR-α, EVR-SF correlations; deduced half life, Eα, production α.
History of 11 EVR-α-SF correlated events (2010Du06,2011Ga19):
Event #1: Energy of the evaporation residue=6.51 MeV: 2010Du06,2011Ga19.
$E_{\alpha 1}$=9982 keV, Δt_1=0.242 s, assigned to ^{288}Fl.
E_{SF}=196 MeV, Δt_2=0.130 s, assigned to ^{284}Cn.
^{284}Cn decays by SF.
Event #2: Energy of the EVR=7.60 MeV: 2010Du06,2011Ga19.
$E_{\alpha 1}$=9814 keV, Δt_1=1.348 s, assigned to ^{288}Fl.
E_{SF}=272 MeV, Δt_2=0.059 s, assigned to ^{284}Cn.
^{284}Cn decays by SF.
Event #3: Energy of the EVR=5.97 MeV: 2010Du06,2011Ga19.
$E_{\alpha 1}$=10062 keV, Δt_1=0.649 s, assigned to ^{288}Fl.
E_{SF}=163 MeV, Δt_2=0.171 s, assigned to ^{284}Cn.
^{284}Cn decays by SF.
Event #4: Energy of the EVR=8.69 MeV: 2010Du06,2011Ga19.
$E_{\alpha 1}$=9915 keV, Δt_1=0.446 s, assigned to ^{288}Fl.
E_{SF}=194 MeV, Δt_2=0.087 s, assigned to ^{284}Cn.
^{284}Cn decays by SF.
Event #5: Energy of the EVR=7.99 MeV: 2010Du06,2011Ga19.
$E_{\alpha 1}$=9919 keV, Δt_1=1.164 s, assigned to ^{288}Fl.
E_{SF}=222 MeV, Δt_2=0.343 s, assigned to ^{284}Cn.
^{284}Cn decays by SF.
Event #6: Energy of the EVR=9.10 MeV: 2010Du06,2011Ga19.
$E_{\alpha 1}$=9980 keV, Δt_1=0.242 s, assigned to ^{288}Fl.
E_{SF}=180 MeV, Δt_2=0.026 s, assigned to ^{284}Cn.
^{284}Cn decays by SF.
Event #7: Energy of the EVR=13.22 MeV: 2010Du06,2011Ga19.
$E_{\alpha 1}$=9947 keV, Δt_1=0.763 s, assigned to ^{288}Fl.
E_{SF}=198 MeV, Δt_2=0.029 s, assigned to ^{284}Cn.
^{284}Cn decays by SF.
Event #8: Energy of the EVR=7.35 MeV: 2010Du06,2011Ga19.
$E_{\alpha 1}$=9700 keV, Δt_1=0.823 s, assigned to ^{288}Fl.
E_{SF}=170 MeV, Δt_2=0.063 s, assigned to ^{284}Cn.
^{284}Cn decays by SF.
Event #9: Energy of the EVR=8.67 MeV: 2010Du06,2011Ga19.
$E_{\alpha 1}$=9965 keV, Δt_1=0.482 s, assigned to ^{288}Fl.
E_{SF}=161 MeV, Δt_2=0.401 s, assigned to ^{284}Cn.
^{284}Cn decays by SF.
Event #10: Energy of the EVR=8.458 MeV: 2011Ga19.
$E_{\alpha 1}$=9947 keV, Δt_1=1.132 s, assigned to ^{288}Fl.
E_{SF}=210 MeV, Δt_2=0.043 s, assigned to ^{284}Cn.
^{284}Cn decays by SF.
Event #11: Energy of the EVR=7.605 MeV: 2011Ga19.
$E_{\alpha 1}$=9848 keV, Δt_1=0.921 s, assigned to ^{288}Fl.
E_{SF}=196 MeV, Δt_2=0.390 s, assigned to ^{284}Cn.
^{284}Cn decays by SF.

2014Ya33: ^{288,289}Fl produced in ^{244}Pu(^{48}Ca,xn),(^{48}Ca,4n) reaction, and separated in-flight using TASCA-COMPACT-GSI separator, with the detection of two atoms, one each assigned to ^{288}Fl and ^{289}Fl. The adsorption behavior of Fl and Cn isotopes was studied together with their homologues Pb, Hg and Rn in groups 14 and 12 using isothermal gas chromatography and thermochromatography techniques. The Fl atoms adsorbed on Au surface formed a metal-metal but indicated higher inertness compared to homologue Pb in the group. It was concluded that Fl is a volatile metal, the least reactive one in group 14, and not as inert as a noble gas.

Event #1: 2014Ya33.

Adopted Levels (continued)

$E_{\alpha 1}$=9.65 MeV, assigned to ^{288}Fl.
E_{SF}=134 MeV, Δt_2=0.650 s, assigned to ^{284}Cn.

For theoretical studies, consult Nuclear Science References (NSR) database at NNDC, BNL for 160 primary references dealing with the half-lives and other aspects of nuclear structure in this mass region.

^{288}Fl Levels

Cross Reference (XREF) Flags

A ^{292}Lv α decay (12.8 ms)

E(level)	J^π	$T_{1/2}$	XREF	Comments
0	0^+	0.64 s *+14−10*	A	%α=100 (2016Ho09); %SF≤3 (2016Ho09) Only the α decay mode has been observed. E(level): the reported activity is assumed to belong to the g.s. of ^{288}Fl. $T_{1/2}$: 644 ms *+138−97* (recommended by 2016Ho09, also 0.66 s *+14−10* in 2015Og05,2017Og01). The recommended value does not include $T_{1/2}$=274 ms *+500−108* from 2017Ka66 (also 2014MoZV), however its inclusion will not affect the adopted value significantly. Others: 0.52 s *+22−13* (2011Ga19, earlier value was 0.47 s *+24−12* in 2010Du06); 0.80 s *+32−18* (2004Og12,2005Og03,2011Og07; 0.63 s *+27−14* in 2004Og07). Theoretical Q(α)=9.1 MeV, and α-decay half-life=631 s (1997Mo25). Eα=9.934 MeV *15* (2016Ho09); 9.94 MeV *2* (2017Ka66); 9.93 MeV *3* (2015Og05,2017Og01,2011Ga19); 9.95 MeV *7* (2004Og12,2004Og07,2005Og03,2011Og07) from ^{288}Fl α decay. E(level): no x-rays or γ-rays were observed in prompt coincidence with any EVR implant or α-particle (2011Ga19). Maximum production σ=9.8 pb *+39−31* (2010Du06,2011Ga19), 5.3 pb *+36−21* (2004Og07).